

Looping Structures

while loops and for loops

Maze

In order to get out of this maze

```
1      vvvv
2  █  (  █
3  █      █
4  █      █
5  █      █
6  █      █
7      vvvv
```

And given the function `move_down()` which moves the character `( )` down 1,

What would the program look like?

While loops

A `while` loop is a style of loop present in most languages which simply act as an `if` statement

```
1  raining = True
2  while raining == True:
3      sky_color = look_at_sky()
4
5      if sky_color == "gray":
6          wait(5)
7      elif sky_color == "blue":
8          get_umbrella()
9          raining = False
10     else:
11         wait(5)
12
13  go_outside()
```

While loop examples

Counting up from 1 to 5

```
1 count = 1
2 while count ≤ 5:
3     print(count)
4     count = count + 1
```

While loop examples

Counting down from 10 to 1

```
1  count = 10
2  while count > 0:
3      print(count)
4      count = count - 1
```

While loop examples

Summing numbers until the user enters 0

```
1 total = 0
2 number = -1
3 while number != 0:
4     number = int(input("Enter a number (0 to stop): "))
5     total = total + number
6
7 print("Total:", total)
```

While loop examples

Guessing game

```
1  secret_number = 7
2  guess = 0
3  while guess != secret_number:
4      guess = int(input("Guess the number: "))
5
6      if guess < secret_number:
7          print("Too low!")
8      elif guess > secret_number:
9          print("Too high!")
10
11 print("You got it!")
```

While loop examples

Doubling a value until it passes a limit

```
1 value = 1
2 while value < 1000:
3     value = value * 2
4     print(value)
```

Break and Continue Statements

In a while loop it's often useful to be able to *stop*, fully before ending

For example,

```
1  while True:
2      answer = input("Type 'quit' to stop: ")
3
4      if answer == "quit":
5          break
```

Even if the [condition] inside the while loop is `True`, we can still get out of the while loop through `break`

While loop with break examples

Searching a list for a target value

```
1  numbers = [4, 8, 15, 16, 23, 42]
2  target = 16
3  found = False
4
5  index = 0
6  while index < len(numbers):
7      if numbers[index] == target:
8          found = True
9          break
10     index = index + 1
11
12  print("Found:", found)
```

While loop with break examples

Validating user input

```
1  while True:
2      age = int(input("Enter your age: "))
3
4      if age >= 0 and age <= 120:
5          break
6
7      print("That's not a valid age, try again")
8
9  print("Thanks, your age is", age)
```

While loop with break examples

Stopping a game loop when a player loses

```
1  health = 100
2
3  while True:
4      health = health - take_damage()
5
6      if health <= 0:
7          print("Game over!")
8          break
9
10     print("Health remaining:", health)
```

While loop with break examples

Stopping as soon as a condition is met, using an "infinite" loop

```
1  number = 1
2
3  while True:
4      if number * number > 100:
5          break
6      print(number, "squared is", number * number)
7      number = number + 1
```

Exercises

Using a `while` loop, write a program that:

1. Prints every even number from 2 to 20
2. Asks the user for a password, and keeps asking until they type "secret"
3. Starts at 100 and keeps subtracting 7 until the value is 0 or below, printing each step
4. Adds up numbers from 1 to 50 and prints the final total
5. Asks the user to enter numbers one at a time, stopping when they type a negative number, and prints how many numbers they entered

Exercises

1. A bank account starts with \$50 and earns \$5 interest every year. Print the balance each year until it reaches \$100. How many years did it take?
2. A car has 12 litres of fuel and uses 3 litres per hour of driving. Print the remaining fuel after each hour until it runs out.
3. A student needs 60 hours of volunteering. They volunteer 4 hours a week. Print how many weeks it takes, and the total hours after each week.
4. You are given a number by the user. Keep dividing it by 2 (using integer division) and print each result, stopping once it reaches 0. Count how many times you divided.
5. A plant is 2cm tall and grows 3cm every day. Print the height each day until it is taller than 30cm.